This completion procedure gives us the single reduction rule $g(x) \to x$. As we observed in the opening example, this reduction rule can be used to evaluate any expression of the form $g^k(x)$ to obtain the value x.

Now let's look at a classic example of a completion procedure from Knuth and Bendix [1970]. It starts with the following three equations, which can be used as axioms for a group, where e is the identity:

$$\begin{aligned} &e \cdot x \leftrightarrow x, \\ &x^{-1} \cdot x \leftrightarrow e, \\ &(x \cdot y) \cdot z \leftrightarrow x \cdot (y \cdot z). \end{aligned} \tag{8.13}$$

The ordering $\succ$ defined on the expressions of this group is a bit complicated. But we'll describe it because it is a nice example of a recursive definition. For an expression s, let

$$n(x, s)$$

denote the number of occurrences of x in s. For an expression s, let $w(s)$ be the sum of $n(e, s)$ and all $n(x, s)$ for variables x in s. For example, if

$$s = (x \cdot e) \cdot (x \cdot y),$$

then

$$w(s) = n(e, s) + n(x, s) + n(y, s) = 1 + 2 + 1 = 4.$$

If s and t are expressions over the group defined by axioms (8.13), then we define $s \succ t$ if either of the following two conditions is satisfied:

1. $w(s) > w(t)$, and $n(s, x) \geq n(t, x)$ for each variable x in t.
2. $w(s) = w(t)$, and $n(s, x) = n(t, x)$ for each variable x in either s or t, and $s \succ t$ matches one of the following patterns, where x^{-n} means that the operation $^{-1}$ is applied n times. For example, $x^{-2} = (x^{-1})^{-1}$.

   $x^{-n} \succ x$ for any variable x and $n \geq 1$.

   $e^{-n} \succ e$ for $n \geq 1$.

   $(a_1 \cdot a_2)^{-1} \succ b_1 \cdot b_2$.

   $a^{-1} \succ b^{-1}$ if $a \succ b$.

   $a_1 \cdot a_2 \succ b_1 \cdot b_2$ if either $a_1 \succ b_1$ or $a_1 = b_1$ and $a_2 \succ b_2$.

For example, condition 1 of the definition tells us that $(x \cdot y) \succ x$. Therefore we can conclude, by condition 2 of the definition, that $(x \cdot y) \cdot z \succ x \cdot (y \cdot z)$.

The inference rules can now be applied to the three equations (8.13) to obtain the following set of reduction rules:

$$
\begin{aligned}
&e \cdot x \to x\\
&x \cdot e \to x\\
&x^{-1} \cdot x \to e\\
&x \cdot x^{-1} \to e\\
&e^{-1} \to e\\
&x^{-1-1} \to x\\
&y^{-1} \cdot (y \cdot z) \to z\\
&y \cdot (y^{-1} \cdot z) \to z\\
&(x \cdot y) \cdot z \to x \cdot (y \cdot z)\\
&(x \cdot y)^{-1} \to y^{-1} \cdot x^{-1}.
\end{aligned}
$$

In the exercises we'll examine how some of these reduction rules are found. These reduction rules statisfy the completion property. In other words, they can be used to reduce two expressions s and t to the same normal form if and only if the equation $s \leftrightarrow t$ is derivable from the three equations (8.13). For example, to see whether the equation $((x \cdot y^{-1}) \cdot z)^{-1} \leftrightarrow (z^{-1} \cdot y) \cdot x^{-1}$ can be derived from equations (8.13), all we need to do is reduce both expressions to normal form as follows:

$$((x \cdot y^{-1}) \cdot z)^{-1} \to z^{-1} \cdot (x \cdot y^{-1})^{-1} \to z^{-1} \cdot (y^{-1-1} \cdot x^{-1}) \to z^{-1} \cdot (y \cdot x^{-1})$$

and

$$(z^{-1} \cdot y) \cdot x^{-1} \to z^{-1} \cdot (y \cdot x^{-1}).$$

Since the normal forms are equal, we have

$$((x \cdot y^{-1}) \cdot z)^{-1} \leftrightarrow (z^{-1} \cdot y) \cdot x^{-1}.$$

We should remark that it's not always possible to obtain a complete set of reduction rules for a system of equations. In other words, the completion procedure may not terminate, or it may terminate with a set of reduction rules that does not generate the same set of equations as the original equations. Finding a complete set of reduction rules depends heavily on the ordering $\succ$ that must be defined on the expressions under consideration. The presentation that we've given here is based on the collection of papers edited by Aït-Kaci and Nivat [1989]. These papers and the original paper by Knuth and Bendix [1970] contain more detailed discussions about ordering expressions. They also include many applications of the procedure.